Unit 5: Turing Machines and Decidability

A Turing machine is $M = (Q, \Sigma, \Gamma, \delta, q0, B, F)$ with an infinite tape, a read/write head and $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$. It accepts if it enters an accepting state.

5.2 Variants

Multi-tape, multi-track and nondeterministic Turing machines all recognise exactly the same languages as the basic single-tape machine; only running time changes.

5.3 The Church–Turing thesis
Every effectively computable function is computable by a Turing machine.

5.4 Undecidability

The halting problem HALT = { <M, w> : M halts on w } is undecidable, proved by diagonalisation.

Rice's theorem: every non-trivial property of recursively enumerable languages is undecidable.

5.5 Post Correspondence Problem

PCP is undecidable and is used to prove that ambiguity of CFGs is undecidable.